Adrian Marin Mag

Curriculum Vitae

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Profile

PhD candidate in geophysics at the University of Oxford, specialising in the development of mathematical and computational methods for linear inverse problems and inference. My work focuses on extracting information from incomplete and noisy data, with particular emphasis on uncertainty quantification using statistical and probabilistic approaches, including Bayesian inversion and confidence interval analysis.
I design and implement computational tools that encode rigorous mathematical models while providing intuitive interfaces for end users. I have experience in scientific Python, optimisation, and numerical modelling, and I am interested in applying these methods to real-world problems in industry.

Core strengths

- **Mathematical modelling & inference:** inverse problems, linear and probabilistic inference (SOLA, Backus–Gilbert), uncertainty quantification, Bayesian methods.
- **Mathematical foundations:** functional analysis, measure theory, abstract linear algebra.
- **Computational methods:** numerical modelling, adjoint-based methods, operator-based formulations, optimisation.
- **Scientific computing:** Python (NumPy/SciPy), packaging (poetry/setuptools), testing (pytest), CI/CD (GitHub Actions), documentation (Sphinx), Linux/HPC (Slurm, MPI basics).

Education

2022–Present **PhD in Deep Earth Imaging**, *University of Oxford*, Oxford
Thesis: *Imaging the multi-scale topography of the Earth's core.*
Advisors: Dr Paula Koelemeijer, Dr Christophe Zaroli, Dr Andrew Walker

2018–2022 **MSci in Geophysics (First-Class Honours, 82%)**, *University College London*, London
Thesis: *Towards building 3D global seismic tomography models using proximal methods*

Research & Software Experience

2022–Present **PhD Researcher (inverse problems & scientific software)**, *University of Oxford*

- Developed methods for **inferring physical structure from incomplete and noisy data** in linear inverse problems.
- Implemented codes for computing **3D seismic sensitivity kernels** using adjoint methods, allowing for more efficient finite frequency tomography.
- Extended SOLA inference method to estimate a wider range of subsurface properties (e.g. gradients, spectral components), and improved **uncertainty quantification** in confidence interval set methods using convex analysis.
- Built and maintained an **open-source Python library** for linear inverse/inference problems, including API design, testing (pytest), documentation (Sphinx), and CI/CD (GitHub Actions).
- Implemented **probabilistic operator-based models** in abstract function spaces, allowing flexible and modular inference workflows.

Publications

2025 Mag, A. M., Zaroli, C., Koelemeijer, P. (2025). Bridging the gap between SOLA and deterministic linear inferences in the context of seismic tomography. *Geophysical Journal International*, 242(1), ggaf131.

Selected Projects

SOLA-PLI (Python) Developed a library for probabilistic linear inferences; packaging, unittesting, docs, examples

SOLA–DLI (Python) Developed a library for linear inferences with deterministic priors; packaging, CI, docs, examples.

Finite Frequency Kernels (Python) Codes for computing volumetric and geometric finite frequency kernels based on adjoint method using AxiSEM3D outputs.

SensRay (Python) Developed simple code for computing ray theoretical sensitvity kernels discertized in 1D and 3D (wrapping taup).

Fast Marching (MATLAB) 2-D wavefront propagation and ray tracing in heterogeneous media for travel-time modelling (numerical methods course project).

Thermo-mechanical Convection (MATLAB) 2-D coupled thermal/viscous flow with temperature- and viscosity-dependence (for fourth year geodynamic undergraduate course project).

Conferences & Presentations

2023 **Poster**: Mapping the Topography of Earth's Core. PGRiP, Edinburgh.

2024 **Poster**: Constraining Earth model properties through Backus–Gilbert SOLA inferences. UKSEDI, Leeds; BSM, Reading.

2025 **Poster**: Combining SOLA and Deterministic Linear Inferences. Inge Lehmann Symposium, Copenhagen.

2025 **Poster**: SOLA inferences and normal modes. MODES/Deep Earth Meeting, Oxford–Cambridge.

2025 **Talk**: Beyond inversions: the flexibility of linear inference methods. PGRiP.

Teaching

2023–2026 **College Tutor (Mathematics)**, *Exeter & Worcester Colleges, University of Oxford*
First- and third-year tutorials: calculus, statistics, vector calculus, continuum mechanics.

2023/2024 **Python Course Demonstrator**, *University of Oxford*
Helped with demonstrating a two-day intermediate scientific Python workshop (NumPy/SciPy/Matplotlib, vscode, linux).

2020–2021 **Mathematics & Physics Tutor**, *Notebook Tutors*, London
Online tutoring for IB/A-level/AP syllabi.

2020-2021 **UCL connected learning intern**, *UCL*, London
Created interactive MATLAB notebooks for a numerical methods module to improve the online delivery of classes.

PGTA **2021-2021**, *UCL*, London
Developed a randomized online question bank for the Global Geophysics exam, including MATLAB scripts to generate unique datasets and solutions.

Technical Skills

Programming	Python (scientific stack), C++ (basic), (basic)
Software	Git, GitHub Actions, pytest, Sphinx, Jupyter, VSCode, Github Copilot, LaTeX,
Numerics	Adjoints, finite-difference/element basics, inverse problems, optimisation
Maths	Calculus, linear algebra, abstract linear algebra, real analysis, introductory topology, introductory functional analysis, ODEs/PDEs, introductory measure theory, introductory abstract probability theory.

Honours & Awards

2023	Grant for SPIN workshop
2022	Travel grant for SEDI conference

References

Dr Paula Koelemeijer	Associate Professor, Department of Earth Sciences, University of Oxford – paula.koelemeijer@earth.ox.ac.uk
Dr Christophe Zaroli	Lecturer, ITES, University of Strasbourg – c.zaroli@unistra.fr